

1. The set of propositions is consistent because there exists an assignment of truth values to the propositional variables that makes all the propositions true.
 1. The system is working if and only if it is getting enough voltage.
 W = The system is working, V = The system is getting enough voltage
 $W \leftrightarrow V$
 2. The system is getting enough voltage, or the system is not connected to a battery.
 V = The system is getting enough voltage, B = The system is connected to a battery
 $V \vee \neg B$ (inclusive "or" here because the system needs enough voltage to work so V should always be true if the system is to work)
 3. If the system is not connected to a battery, then the battery is charged.
 B = The system is connected to a battery, C = The battery is charged
 $\neg B \rightarrow C$
 4. If the system is not getting enough voltage, then the battery is not charged.
 V = The system is getting enough voltage, C = The battery is charged
 $\neg V \rightarrow \neg C$
 5. The battery is charged.
 C = The battery is charged
 C

To see if all these propositions are consistent, the following proposition must evaluate to true:

$$C \wedge (W \leftrightarrow V) \wedge (V \vee \neg B) \wedge (\neg B \rightarrow C) \wedge (\neg V \rightarrow \neg C)$$

- For all to be true, the first C must be true so all occurrences of C must be true and therefore all occurrences of $\neg C$ must be false.
- For $(W \leftrightarrow V)$ to be true both W and V need to be either true or false, for this proof they will both be true.
- Now all occurrences of V will be considered true and all occurrences of $\neg V$ will be false.
- For $(V \vee \neg B)$ to be true either V or $\neg B$ need to be true. Since V is already true $\neg B$ can be either true or false to make this proposition true.
- Because C must be true that means that $(\neg B \rightarrow C)$ is " $\neg B$ implies a true value". Since both a true value or a false value can imply a true value for this proposition to be true $\neg B$ can be either true or false.
- For proving these propositions to be consistent, knowing $(V \vee \neg B)$ and $(\neg B \rightarrow C)$ can have $\neg B$ be either true or false, it will be denoted as true.
- For $(\neg V \rightarrow \neg C)$, $\neg V$ is false and so is $\neg C$ and false implies false leads to a true value, so this proposition is also true.

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$C \wedge (W \leftrightarrow V) \wedge (V \vee \neg B) \wedge (\neg B \rightarrow C) \wedge (\neg V \rightarrow \neg C)$ is made true.

Since there is an assignment of truth values to each of the propositional variables, that makes all propositions true, then this set of propositions is consistent.